Massey University
College of Sciences

161.304: Comments and Solutions for Computer Laboratory 2

Exercise 1: Some probability questions

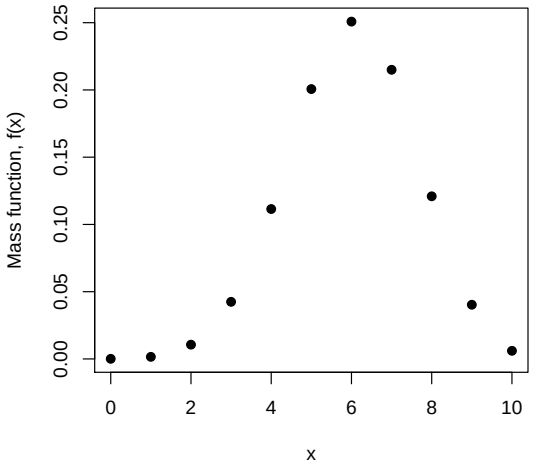
- 2E1 $\rightarrow$ (2) and (4)
- 2E2 $\rightarrow$ (3)
- 2E3 $\rightarrow$ (1) and (4) and (4) is Bayes formula :)
- 2E4 This statement is per se a subjective truth that is very likely untrue at any (most) moment(s) in time, quite approximate, and cannot be exactly verified. Yet, it is a useful approximate statement.

Exercise 2: Probability Calculations
The code below provides the required probabilities.

1. `dpois(0,lambda=3)` returns 0.04978707.
2. `dpois(2,lambda=4.321)` returns 0.1240372.
3. `ppois(4,lambda=3)` returns 0.8152632.
4. `ppois(3,lambda=3)` returns 0.6472319. Note the $\mathsf{P}(X < 4) = \mathsf{P}(X \leq 3)$ for a Poisson random variable X, since it only takes non-negative integer values.
5. `ppois(4,lambda=3) - ppois(2,lambda=3)` returns 0.3920732.
6. `pnorm(5,mean=4,sd=1.5)` returns 0.7475075.
7. `1 - pnorm(5,mean=4,sd=1.5)` returns 0.2524925. We use here the fact that $\mathsf{P}(X > 5) = 1 - \mathsf{P}(X \leq 5)$.
8. `pnorm(6.5,mean=4,sd=1.5) - pnorm(2.5,mean=4,sd=1.5)` returns 0.7935544.
9. `pexp(1.4,rate=1.2)` returns 0.813626.
10. `pgamma(6.54,shape=3,rate=1.1) - pgamma(3.21,shape=3,rate=1.1)` returns 0.2895617.

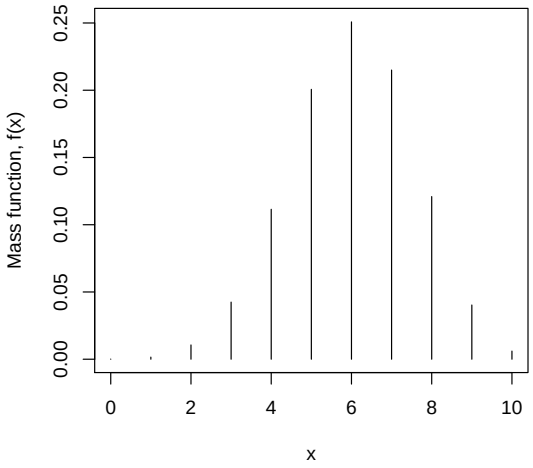
Exercise 3: Plotting Binomial Probability Mass Functions The plot below is produced by the following code.

```
x <- 0:10
f <- dbinom(x,size=10,prob=0.6)
plot(x,f,pch=19,ylab="Mass function, f(x)")
```



Or replacing the last line:

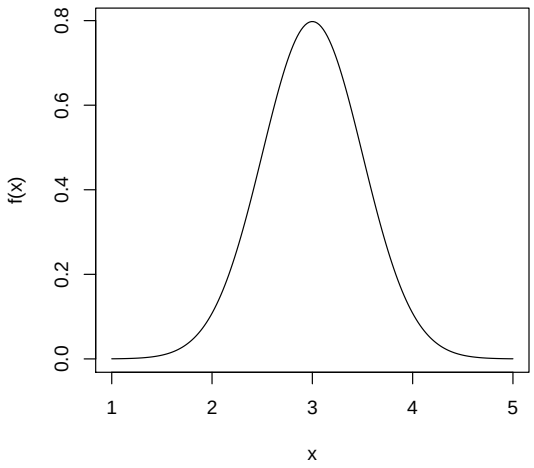
```
plot(x,f,ylab="Mass function, f(x)",type="h")
```



Exercise 4: Plotting Density Functions

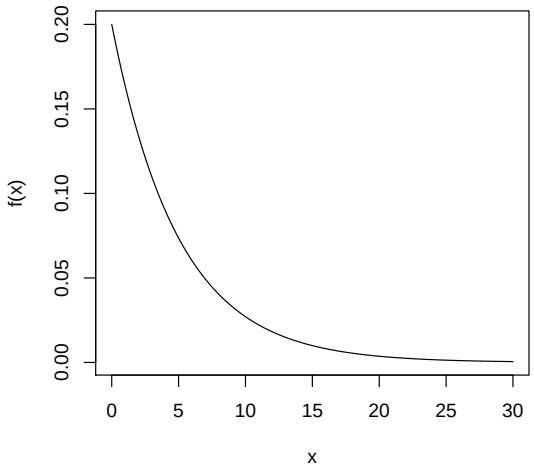
1. The plot below is produced by the following code.

```
x <- seq(1,5,length=200)
f <- dnorm(x,mean=3,sd=0.5)
plot(x,f,type="l",ylab="f(x)")
```



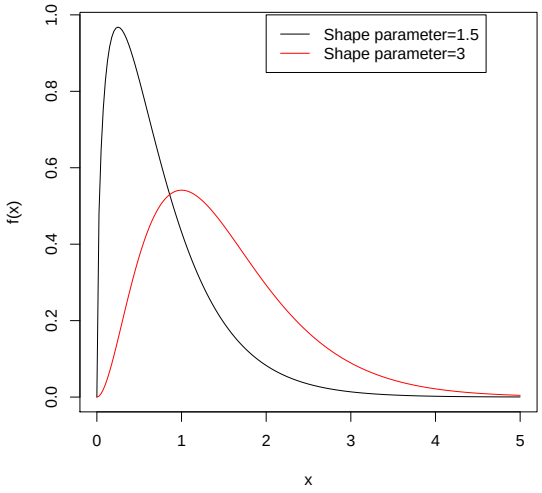
2. The plot below is produced by the following code.

```
x <- seq(0,30,length=200)
f <- dexp(x,rate=0.2)
plot(x,f,type="l",ylab="f(x)")
```



3. The plot below is produced by the following code.

```
x <- seq(0,5,length=200)
f1 <- dgamma(x,shape=1.5,rate=2)
f2 <- dgamma(x,shape=3,rate=2)
plot(x,f1,type="l",ylab="f(x)")
lines(x,f2,col=2)
legend(2,1,lty=c(1,1),col=c(1,2),legend=c("Shape parameter=1.5","Shape parameter=3"))
```


Shape parameter=1.5
Shape parameter=3
f(x)
x
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5